

AI AGENT

Day 92
10/10/2025

An AI Agent is an autonomous System (Software program) that perceives its environment, makes decisions, and acts to achieve specific goals - often using AI techniques like Machine learning, NLP or reasoning

In Simple Words

An Agent thinks, decides, and acts - without human intervention to solve a problem

Example Analogy

Imagine a Self driving car (Car's software system is an AI Agent)

- Perceives : Sensors detect traffic, obstacles, and signs
- Thinks : The onboard AI decides how to steer, accelerate or brake
- Acts : It controls the car safely to reach the destination

Why We Need AI Agents

Reason

Description

Automation

To reduce human effort by making system act intelligently

Efficiency

Agents can process huge data faster than humans

Adaptability

Agents can learn and improve through feedback

Continuous Operation

Can run 24x7 without fatigue

Decision-making

Useful in uncertain or complex environments (like trading, logistics, or customer support)

🌱 Types of AI Agents

AI Agents are classified based on how they make decisions and learn from their environment.

Type	Description	Example
1. Simple Reflex Agent	Acts only based on the <i>current</i> situation (no memory).	A thermostat that turns off heat when temperature exceeds a limit.
2. Model-Based Reflex Agent	Uses internal memory (a model of the world) to decide.	A robot vacuum remembering room layout.
3. Goal-Based Agent	Takes actions that help achieve a <i>specific goal</i> .	GPS choosing the best route to a destination.
4. Utility-Based Agent	Chooses actions that maximize <i>happiness or efficiency</i> .	Stock trading bot maximizing profit.
5. Learning Agent	Learns from experience to improve performance.	ChatGPT, recommendation systems, etc.
6. Multi-Agent System	Multiple agents interact/cooperate/compete.	Autonomous drones coordinating in a swarm.

🌐 Where AI Agents Are Used

Domain	Example
🗣️ Customer Support	Chatbots, virtual assistants (e.g., Siri, ChatGPT)
🚗 Autonomous Vehicles	Self-driving cars, drones
🏠 Finance	Fraud detection, stock trading agents
🏥 Healthcare	Diagnosis systems, medical imaging agents
🏡 Smart Homes	AI assistants controlling devices
🎮 Gaming	Non-player characters (NPCs) with intelligent behavior
⚙️ DevOps / APIs	Agents managing system health and responding to incidents

How to Build An AI Agent

Building an AI Agent involves combining sensing, thinking, and acting capabilities

Step 1: Define the purpose

Decide what problem the agent will solve.

Example: A "Motivational Quote Agent" that fetches and displays quotes daily.

Step 2: Perception Layer (Input)

- Gather data (text, image, voice sensor input, or API input)

Step 3: Decision Layer (Thinking)

- Apply logic, rules, or ML models to analyze the data.
- Techniques used:
 - Rule-based systems
 - Machine learning models
 - Neural networks
 - Reinforcement learning

◆ Step 4: Action Layer (Output)

- Perform an action (send message, control device, call an API, or print result).

◆ Step 5: Learning/Feedback Loop

- Agent improves with experience using reinforcement or supervised learning.

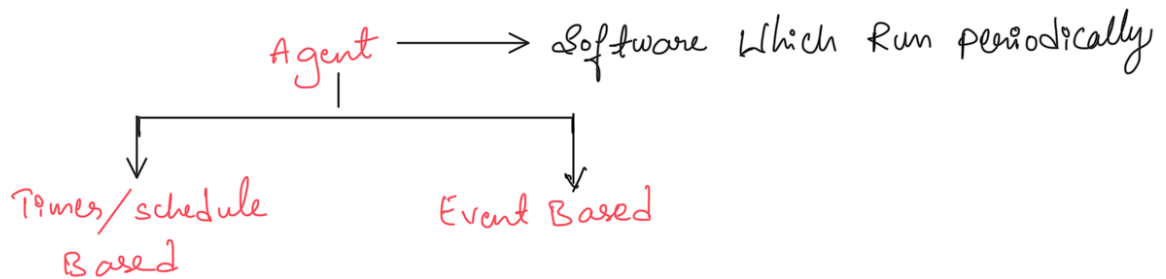
Gen AI → LLM's this LLM's playing a Descriptive Role, a Very good role in making lot of Scenarios Automatable

Suppose I want some task is to be done By Gen AI

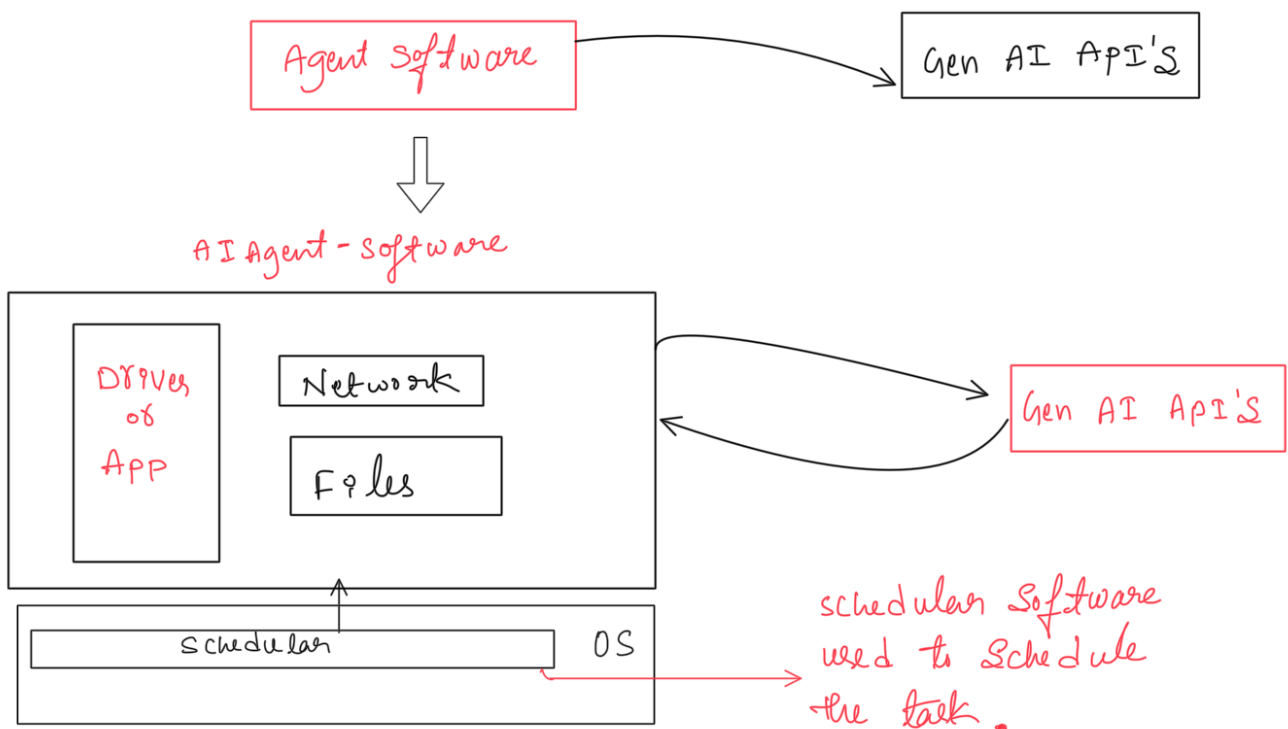
for Example I need a Motivational Quote For that I need to open Website and I have to give the Gen AI prompt saying that give me a Motivational Quote then the gen AI gives me a Quote, If I need Another Quote then Again I have to go to the Same gen AI website & have to repeat the same procedure

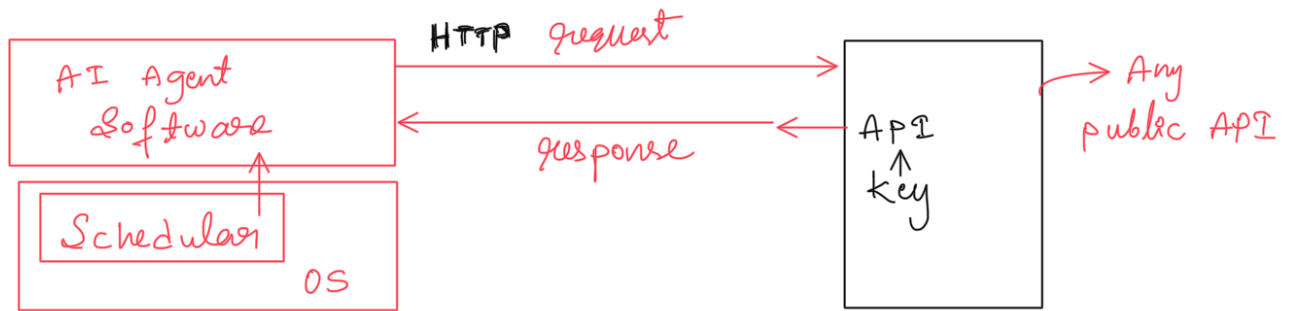
If the work is Repetative and you want it to be done Automatic By computers Without Any manaul Supervision then we use AI Agent

The Agent is a Software code that makes tasks Automated

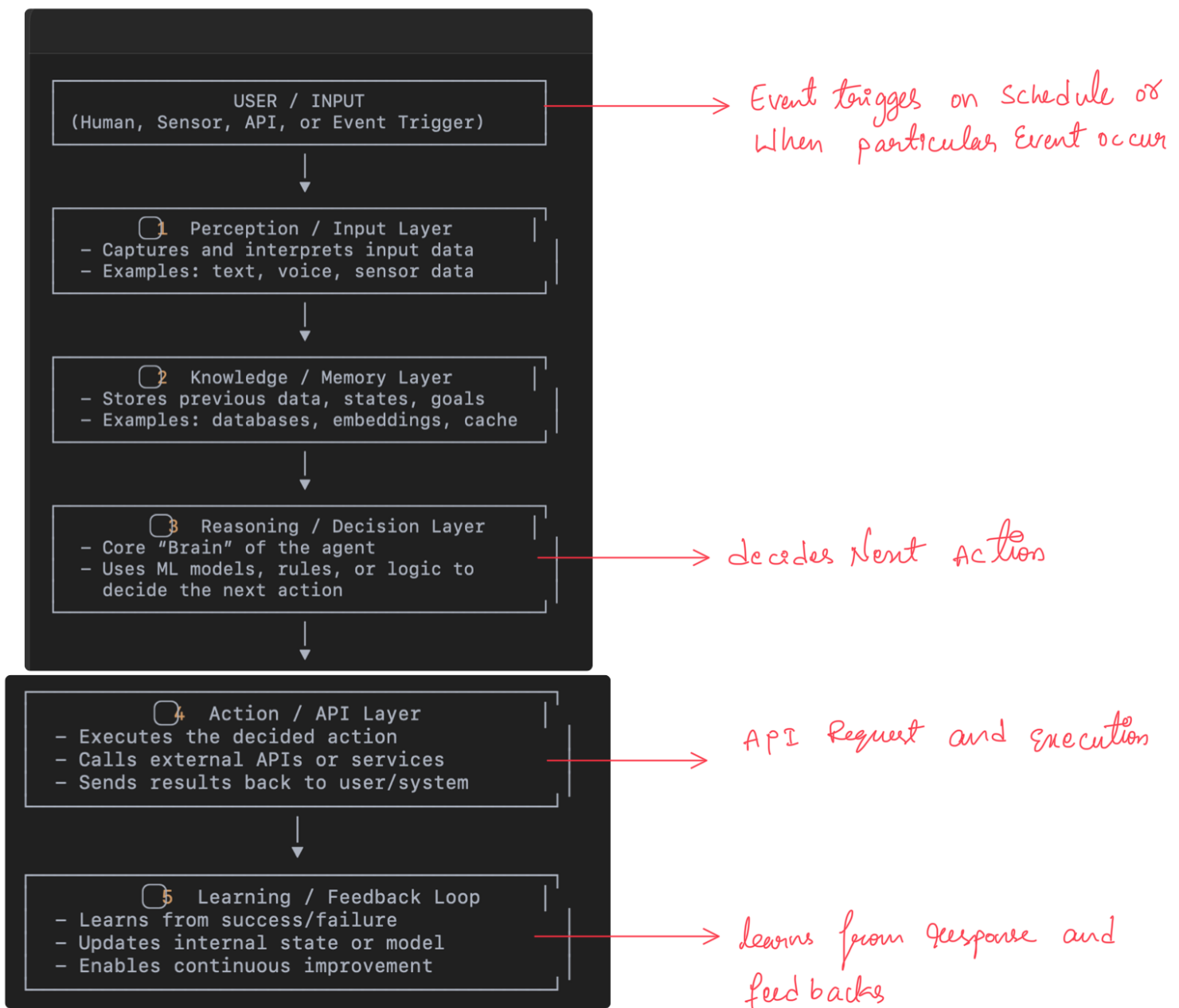


High level design of an AI Agent → Design

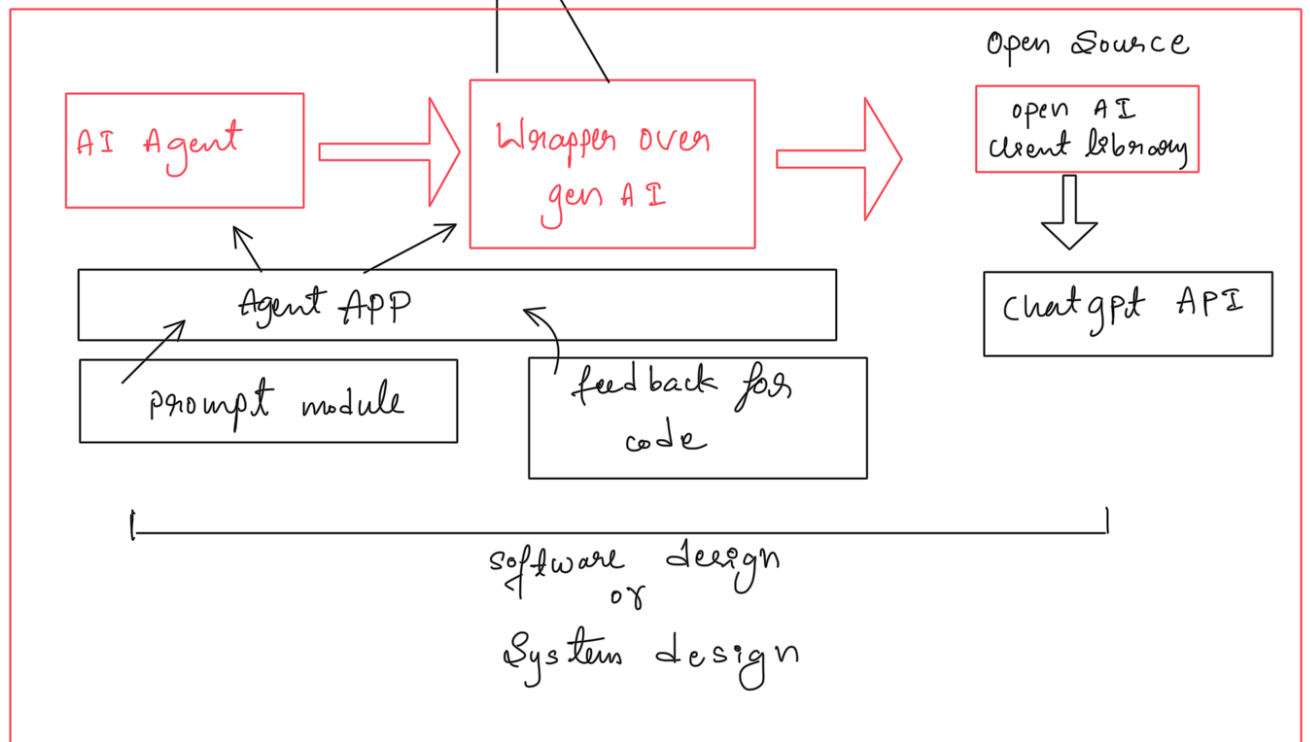
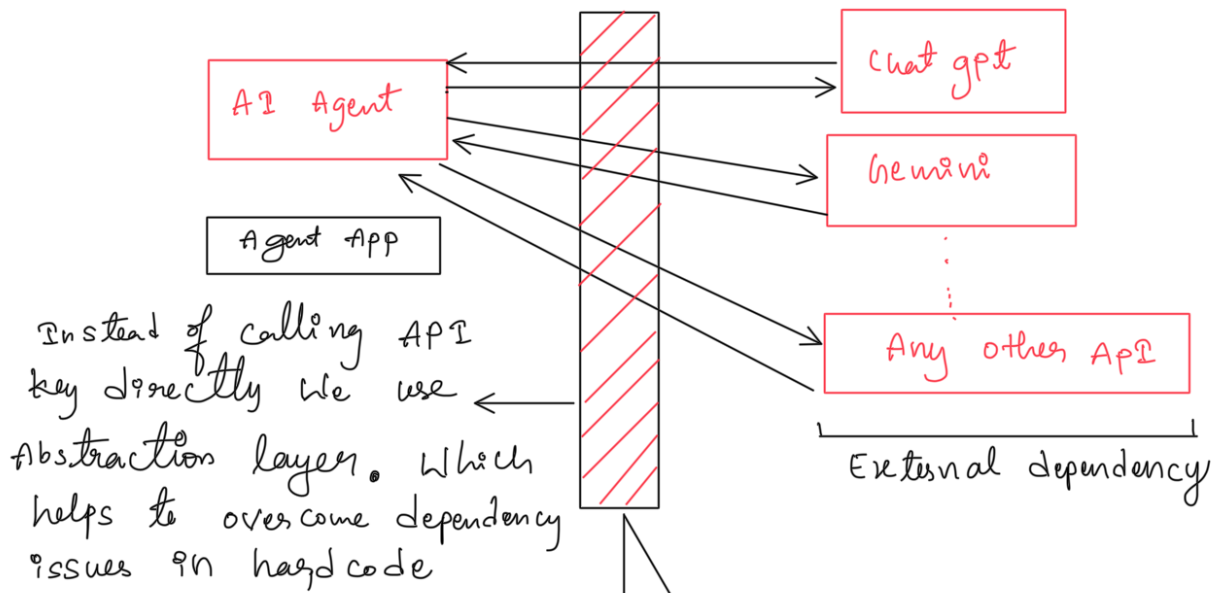




NOTE: NOT All API'S are free to use there is a cost for using API key



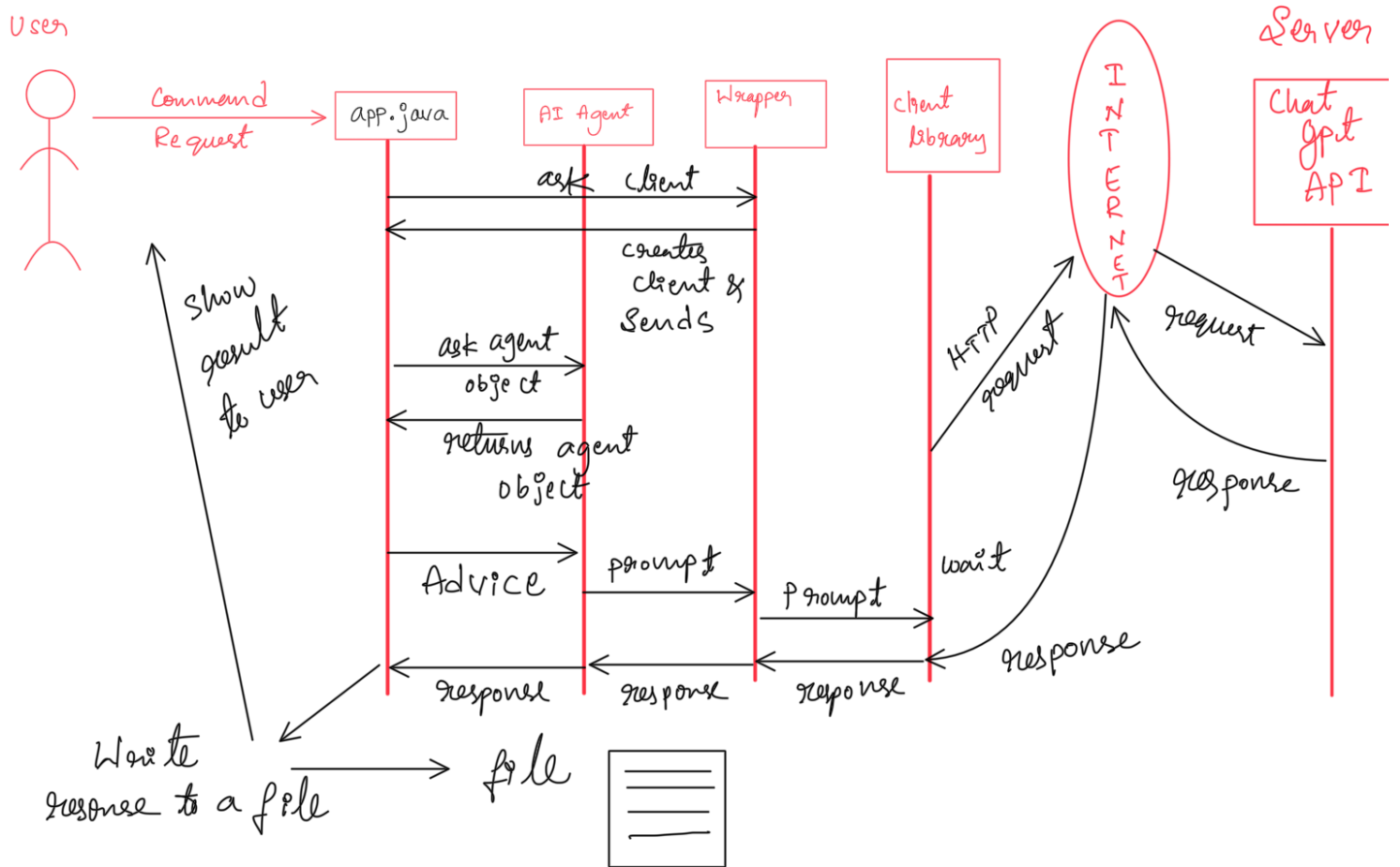
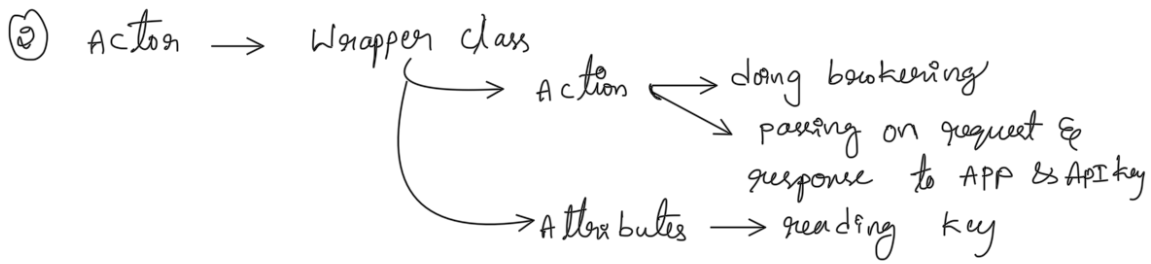
AI Agent can use Any API key's its called External dependency



Modular design

3 A's of Object oriented programming
Actors, Actions, Attributes,

- ① Action → open source open-client-lib
 - Action → request & response
 - Attributes API keys, LLM Model



Work flow of AI Agent Software

